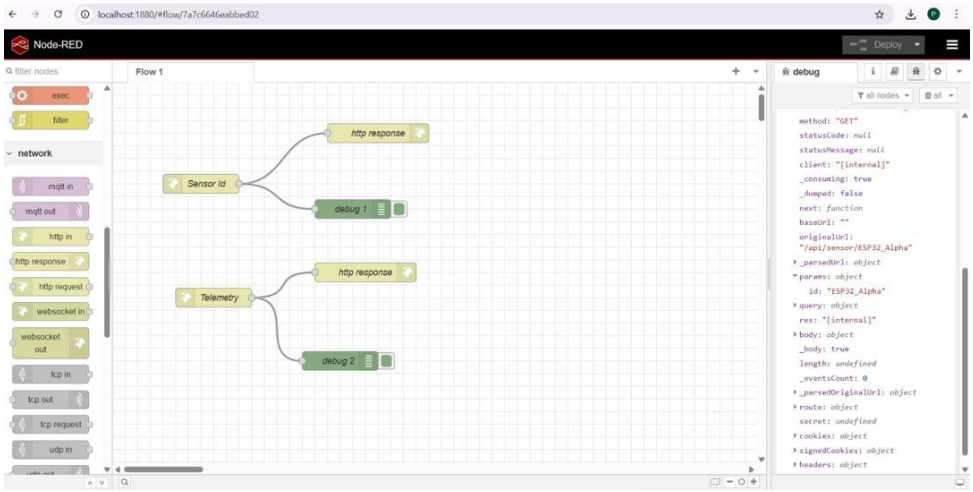
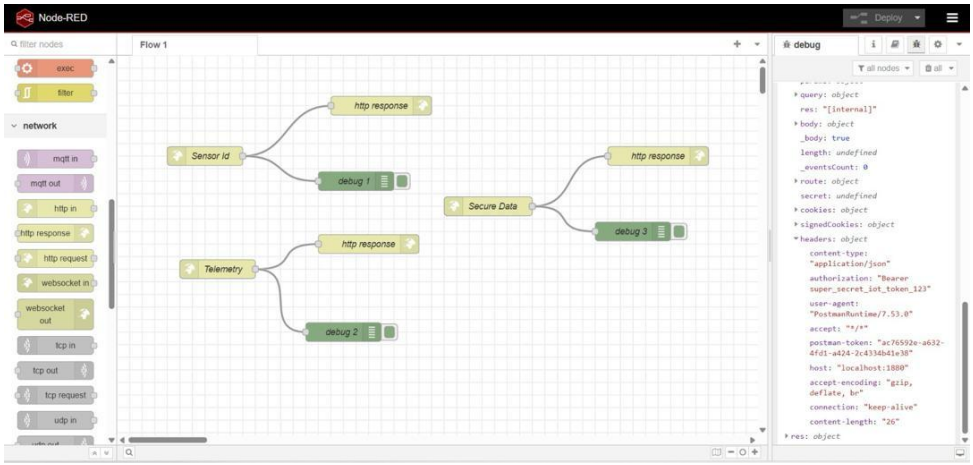
MAHENDRAN A/L THILAGAMANY B122510490

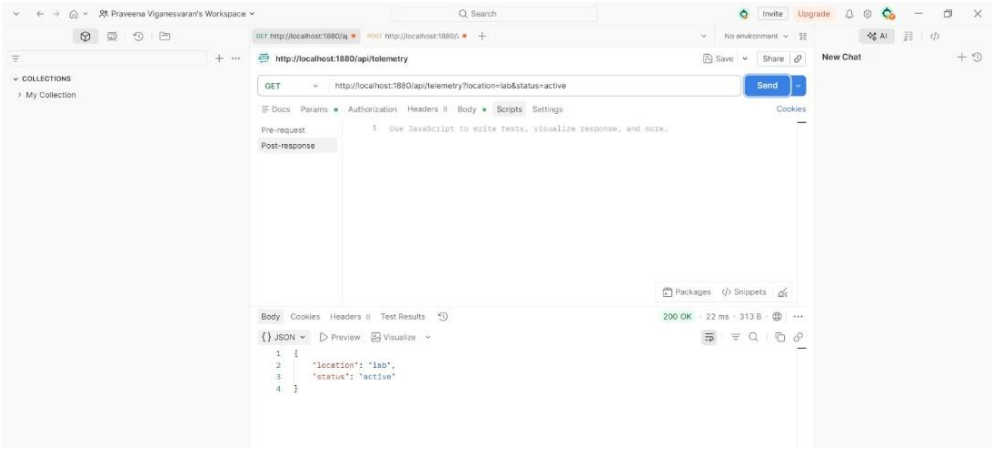
EXERCISE 3: PACKET INSPECTION

Step 1: Query Strings vs. Route Parameters in Node-RED



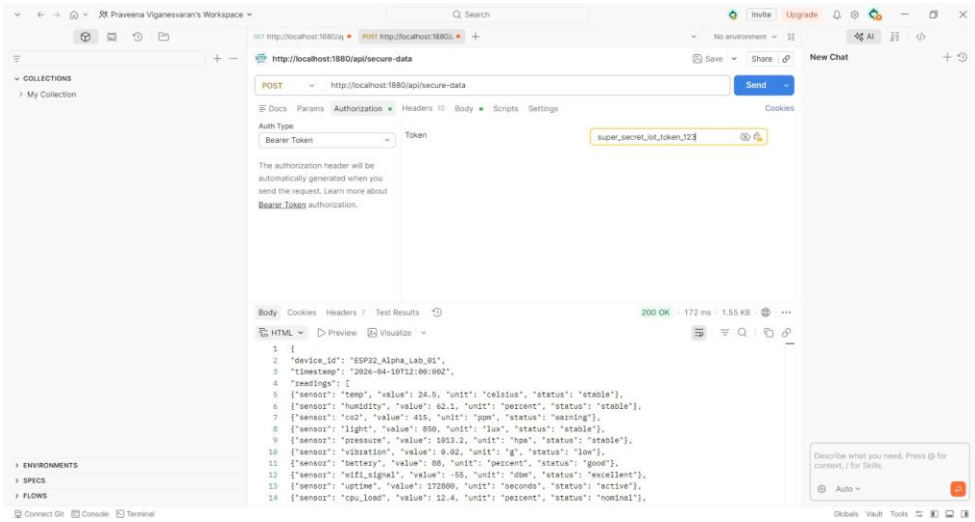


(Node-RED)



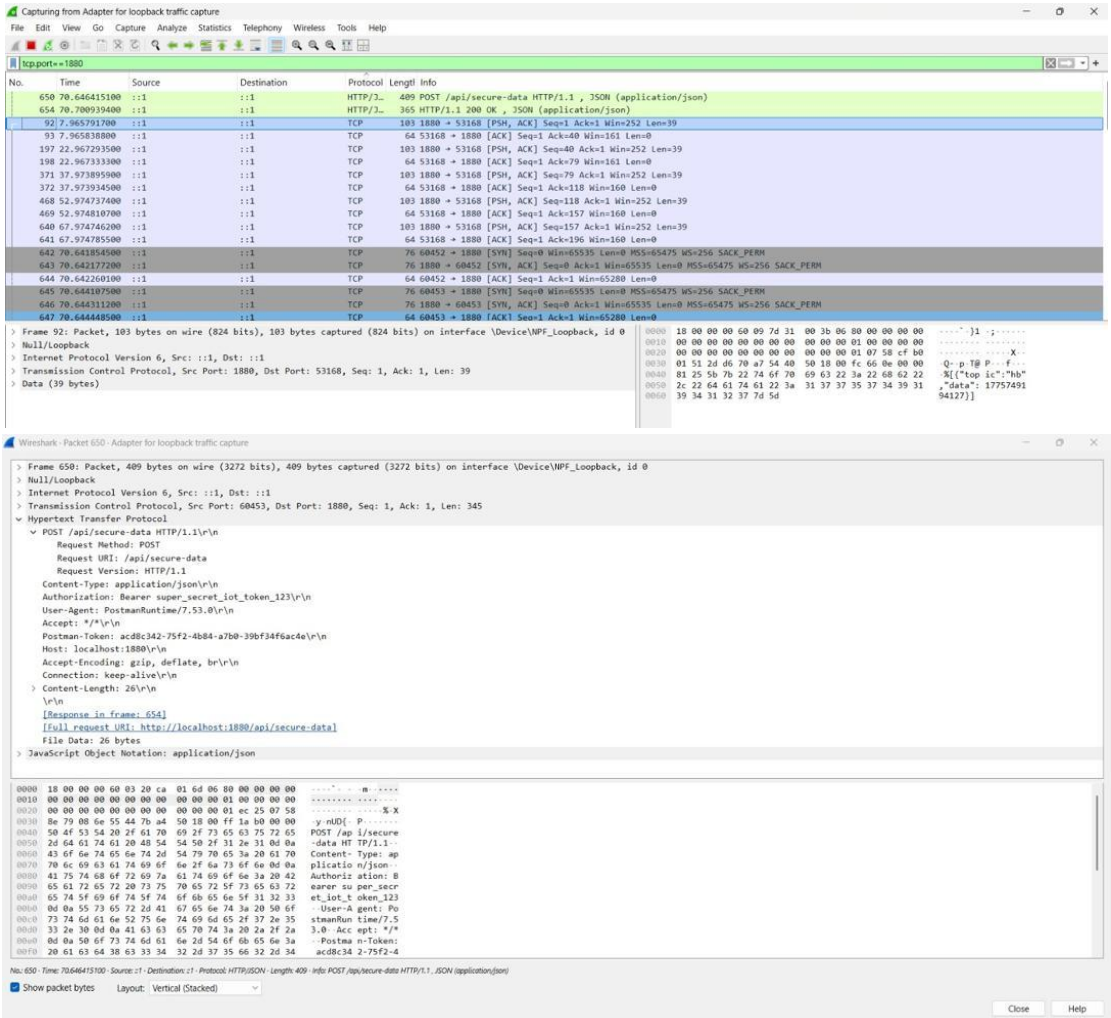
(Postman)

Step 2: The Power of HTTP Headers (Media & Authentication)

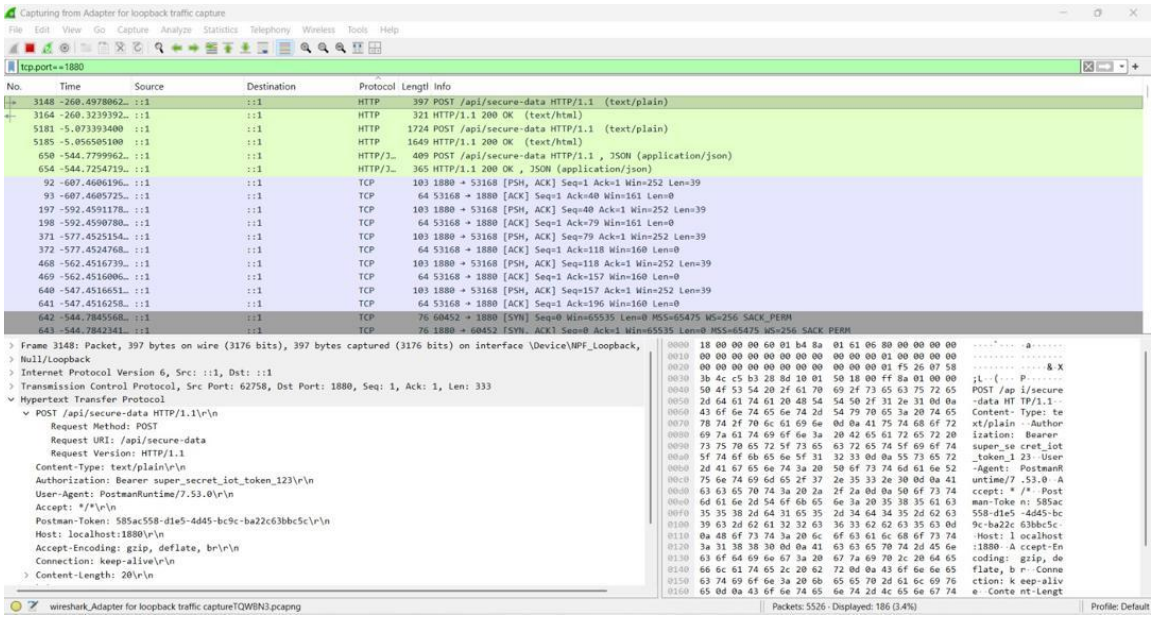


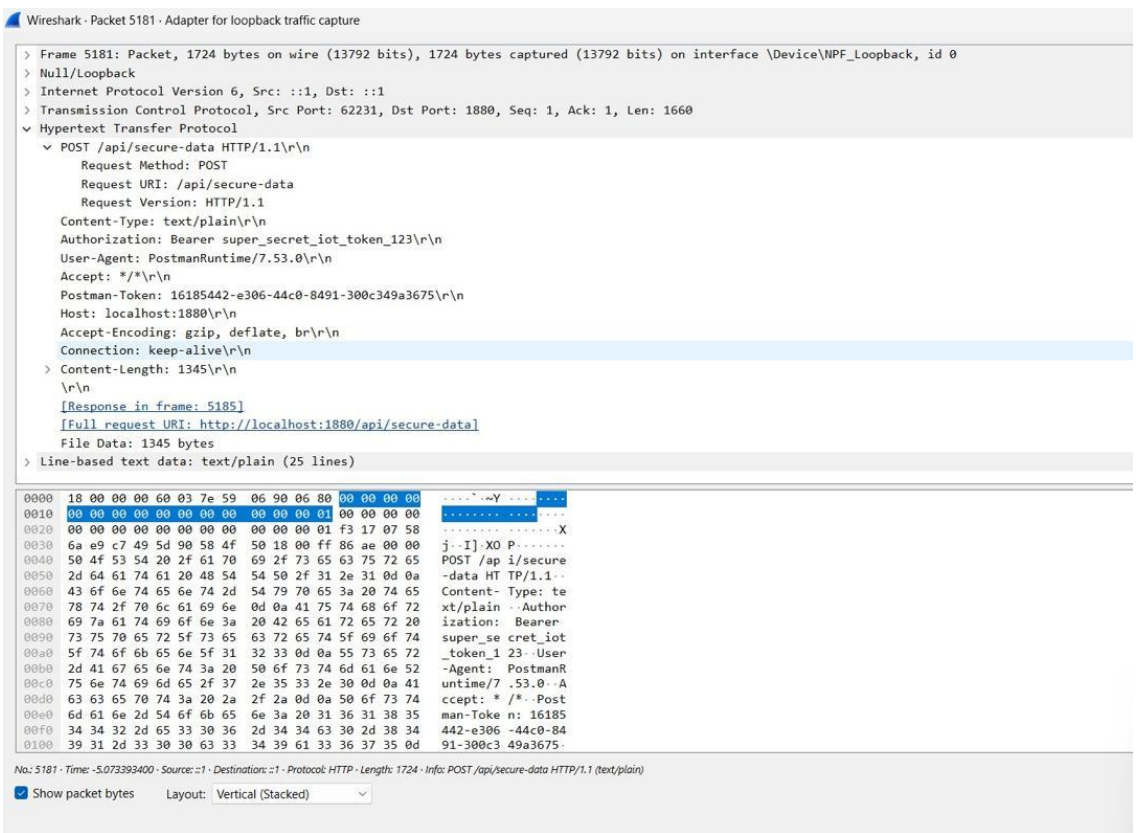
(Postman)

Step 3: Deep Packet Inspection with Wireshark



Part 4: Analytical Modifications & Network Comparison





Wireshark · Packet 650 · Adapter for loopback traffic capture

```
> Frame 650: Packet, 409 bytes on wire (3272 bits), 409 bytes captured (3272 bits) on interface \Device\NPF_Loopback, id 0
> Null/Loopback
> Internet Protocol Version 6, Src: ::1, Dst: ::1
> Transmission Control Protocol, Src Port: 60453, Dst Port: 1880, Seq: 1, Ack: 1, Len: 345
v Hypertext Transfer Protocol
  v POST /api/secure-data HTTP/1.1\r\n
      Request Method: POST
      Request URI: /api/secure-data
      Request Version: HTTP/1.1
    Content-Type: application/json\r\n
    Authorization: Bearer super_secret_iot_token_123\r\n
    User-Agent: PostmanRuntime/7.53.0\r\n
    Accept: */*\r\n
    Postman-Token: acd8c342-75f2-4b84-a7b0-39bf34f6ac4e\r\n
    Host: localhost:1880\r\n
    Accept-Encoding: gzip, deflate, br\r\n
    Connection: keep-alive\r\n
  > Content-Length: 26\r\n
    \r\n
```

```
0030  8e 79 08 6e 55 44 7b a4  50 18 00 ff 1a b0 00 00   ·y·nUD{· P·------
0040  50 4f 53 54 20 2f 61 70  69 2f 73 65 63 75 72 65   POST /ap i/secure
0050  2d 64 61 74 61 20 48 54  54 50 2f 31 2e 31 0d 0a   -data HT TP/1.1··
0060  43 6f 6e 74 65 6e 74 2d  54 79 70 65 3a 20 61 70   Content- Type: ap
0070  70 6c 69 63 61 74 69 6f  6e 2f 6a 73 6f 6e 0d 0a   plicatio n/json··
0080  41 75 74 68 6f 72 69 7a  61 74 69 6f 6e 3a 20 42   Authoriz ation: B
0090  65 61 72 65 72 20 73 75  70 65 72 5f 73 65 63 72   earer su per_secr
00a0  65 74 5f 69 6f 74 5f 74  6f 6b 65 6e 5f 31 32 33   et_iot_t oken_123
00b0  0d 0a 55 73 65 72 2d 41  67 65 6e 74 3a 20 50 6f   ··User-A gent: Po
00c0  73 74 6d 61 6e 52 75 6e  74 69 6d 65 2f 37 2e 35   stmanRun time/7.5
00d0  33 2e 30 0d 0a 41 63 63  65 70 74 3a 20 2a 2f 2a   3.0··Acc ept: */*
00e0  0d 0a 50 6f 73 74 6d 61  6e 2d 54 6f 6b 65 6e 3a   ··Postma n-Token:
00f0  20 61 63 64 38 63 33 34  32 2d 37 35 66 32 2d 34    acd8c34 2-75f2-4
0100  62 38 34 2d 61 37 62 30  2d 33 39 62 66 33 34 66   b84-a7b0 -39bf34f
0110  36 61 63 34 65 0d 0a 48  6f 73 74 3a 20 6c 6f 63   6ac4e··H ost: loc
0120  61 6c 68 6f 73 74 3a 31  38 38 30 0d 0a 41 63 63   alhost:1 880··Acc
0130  65 70 74 2d 45 6e 63 6f  64 69 6e 67 3a 20 67 7a   ept-Enco ding: gz
0140  69 70 2c 20 64 65 66 6c  61 74 65 2c 20 62 72 0d   ip, defl ate, br·
0150  0a 43 6f 6e 6e 65 63 74  69 6f 6e 3a 20 6b 65 65   ·Connect ion: kee
0160  70 2d 61 6c 69 76 65 0d  0a 43 6f 6e 74 65 6e 74   p-alive· ·Content
0170  2d 4c 65 6e 67 74 68 3a  20 32 36 0d 0a 0d 0a 20   -Length:  26····
0180  7b 22 73 74 61 74 75 73  22 3a 20 22 73 65 63 75   {"status ": "secu
0190  72 65 20 74 65 73 74 22  7d                        re test" }
```

HTTP Content-Type header (http.content_type), 32 bytes

Show packet bytes Layout: Vertical (Stacked)

Wireshark · Packet 5181 · Adapter for loopback traffic capture

```
> Transmission Control Protocol, Src Port: 62231, Dst Port: 1880, Seq: 1, Ack: 1, Len: 1660
v Hypertext Transfer Protocol
  v POST /api/secure-data HTTP/1.1\r\n
      Request Method: POST
      Request URI: /api/secure-data
      Request Version: HTTP/1.1
    Content-Type: text/plain\r\n
    Authorization: Bearer super_secret_iot_token_123\r\n
    User-Agent: PostmanRuntime/7.53.0\r\n
    Accept: */*\r\n
    Postman-Token: 16185442-e306-44c0-8491-300c349a3675\r\n
    Host: localhost:1880\r\n
    Accept-Encoding: gzip, deflate, br\r\n
    Connection: keep-alive\r\n
  > Content-Length: 1345\r\n
    \r\n
    [Response in frame: 5185]
    [Full request URI: http://localhost:1880/api/secure-data]
    File Data: 1345 bytes
```

```
0060  43 6f 6e 74 65 6e 74 2d  54 79 70 65 3a 20 74 65   Content- Type: te
0070  78 74 2f 70 6c 61 69 6e  0d 0a 41 75 74 68 6f 72   xt/plain ··Author
0080  69 7a 61 74 69 6f 6e 3a  20 42 65 61 72 65 72 20   ization:  Bearer
0090  73 75 70 65 72 5f 73 65  63 72 65 74 5f 69 6f 74   super_se cret_iot
00a0  5f 74 6f 6b 65 6e 5f 31  32 33 0d 0a 55 73 65 72   _token_1 23··User
00b0  2d 41 67 65 6e 74 3a 20  50 6f 73 74 6d 61 6e 52   -Agent:  PostmanR
00c0  75 6e 74 69 6d 65 2f 37  2e 35 33 2e 30 0d 0a 41   untime/7 .53.0··A
00d0  63 63 65 70 74 3a 20 2a  2f 2a 0d 0a 50 6f 73 74   ccept: * /*··Post
00e0  6d 61 6e 2d 54 6f 6b 65  6e 3a 20 31 36 31 38 35   man-Toke n: 16185
00f0  34 34 32 2d 65 33 30 36  2d 34 34 63 30 2d 38 34   442-e306 -44c0-84
0100  39 31 2d 33 30 30 63 33  34 39 61 33 36 37 35 0d   91-300c3 49a3675·
0110  0a 48 6f 73 74 3a 20 6c  6f 63 61 6c 68 6f 73 74   ·Host: l ocalhost
0120  3a 31 38 38 30 0d 0a 41  63 63 65 70 74 2d 45 6e   :1880··A ccept-En
0130  63 6f 64 69 6e 67 3a 20  67 7a 69 70 2c 20 64 65   coding:  gzip, de
0140  66 6c 61 74 65 2c 20 62  72 0d 0a 43 6f 6e 6e 65   flate, b r··Conne
0150  63 74 69 6f 6e 3a 20 6b  65 65 70 2d 61 6c 69 76   ction: k eep-aliv
0160  65 0d 0a 43 6f 6e 74 65  6e 74 2d 4c 65 6e 67 74   e··Conte nt-Lengt
0170  68 3a 20 31 33 34 35 0d  0a 0d 0a 7b 0d 0a 20 20   h: 1345· ···{··
0180  22 64 65 76 69 63 65 5f  69 64 22 3a 20 22 45 53   "device_ id": "ES
0190  50 33 32 5f 41 6c 70 68  61 5f 4c 61 62 5f 30 31   P32_Alph a_Lab_01
01a0  22 2c 0d 0a 20 20 22 74  69 6d 65 73 74 61 6d 70   ",··  "t imestamp
01b0  22 3a 20 22 32 30 32 36  2d 30 34 2d 31 30 54 31   ": "2026 -04-10T1
01c0  32 3a 30 30 3a 30 30 5a  22 2c 0d 0a 20 20 22 72   2:00:00Z ",··  "r
```

HTTP Content-Type header (http.content_type), 26 bytes

Show packet bytes Layout: Vertical (Stacked)

(Wireshark)